Text Mining

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Background: Social Media Analysis

- Business problem: **What do people think of "iPhone"?**
- Data: real-time social media feeds, news, ...

Human ability cannot keep up with growing data in real time.

- Suppose one person reads 1,000 documents per day and summarizes them on "iPhone"
- Less than 1% of them mention "iPhone," resulting in fewer than 10 related data per day.
- New data is constantly being created and accumulated while reading and summarizing existing data.

Text Mining: definition

Text Mining (Def. *Wikipedia*)

Text mining, also known as intelligent text analysis, text data mining or knowledge-discovery in text (KDT), refers generally to the process of extracting interesting and non-trivial information and knowledge from unstructured text. Text mining is a young interdisciplinary field which draws on information retrieval, data mining, machine learning, statistics and computational linguistics. As most information (over 80%) is stored as text, text mining is believed to have a high commercial potential value.

Rene Witte, Introduction to Text Mining, Tutorial at EDBT, 2006.

Text Mining Around Us

- Sentiment analysis: Restaurant Reviews on yelp

The food was great! But I didn't like the service.

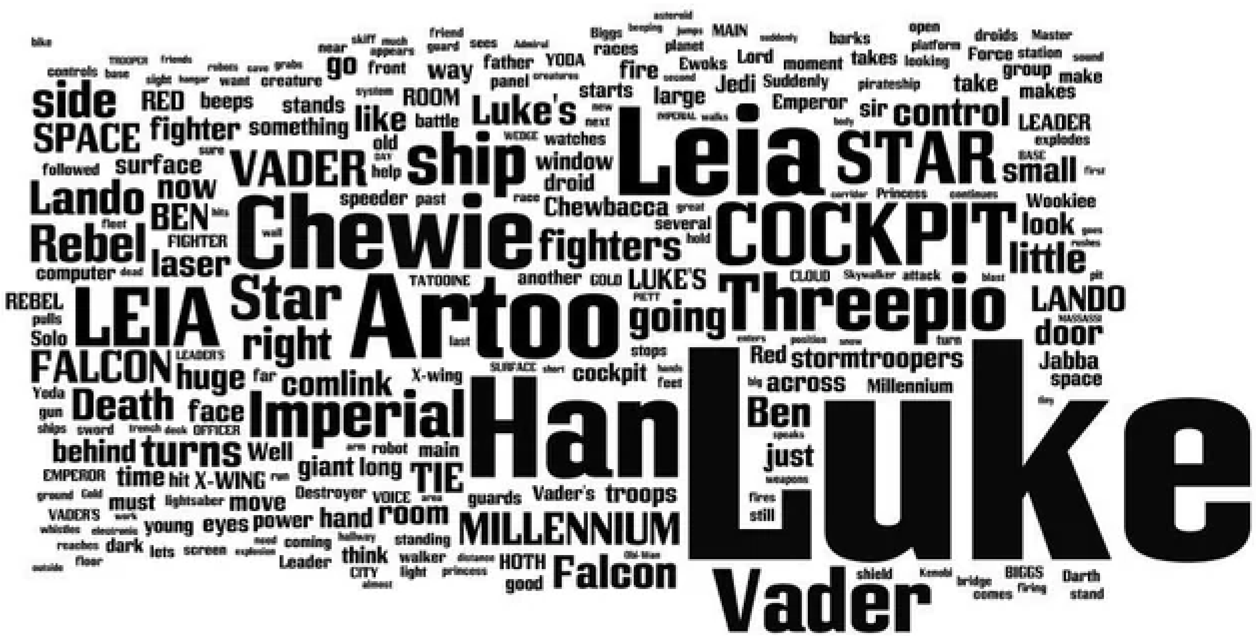
I will definitely come again. Great menu.

The atmosphere is nice, and the service was helpful. When it comes to food, I would say 7/10.

Not my style, I don't recommend it.

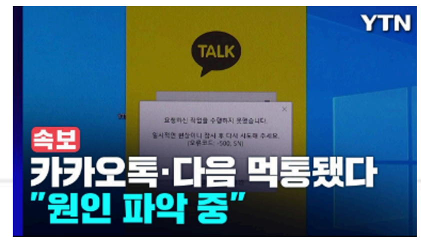
Text Mining Around Us

- Document Summarization
  - a word cloud about "Star Wars," made with the scripts of the scripts.



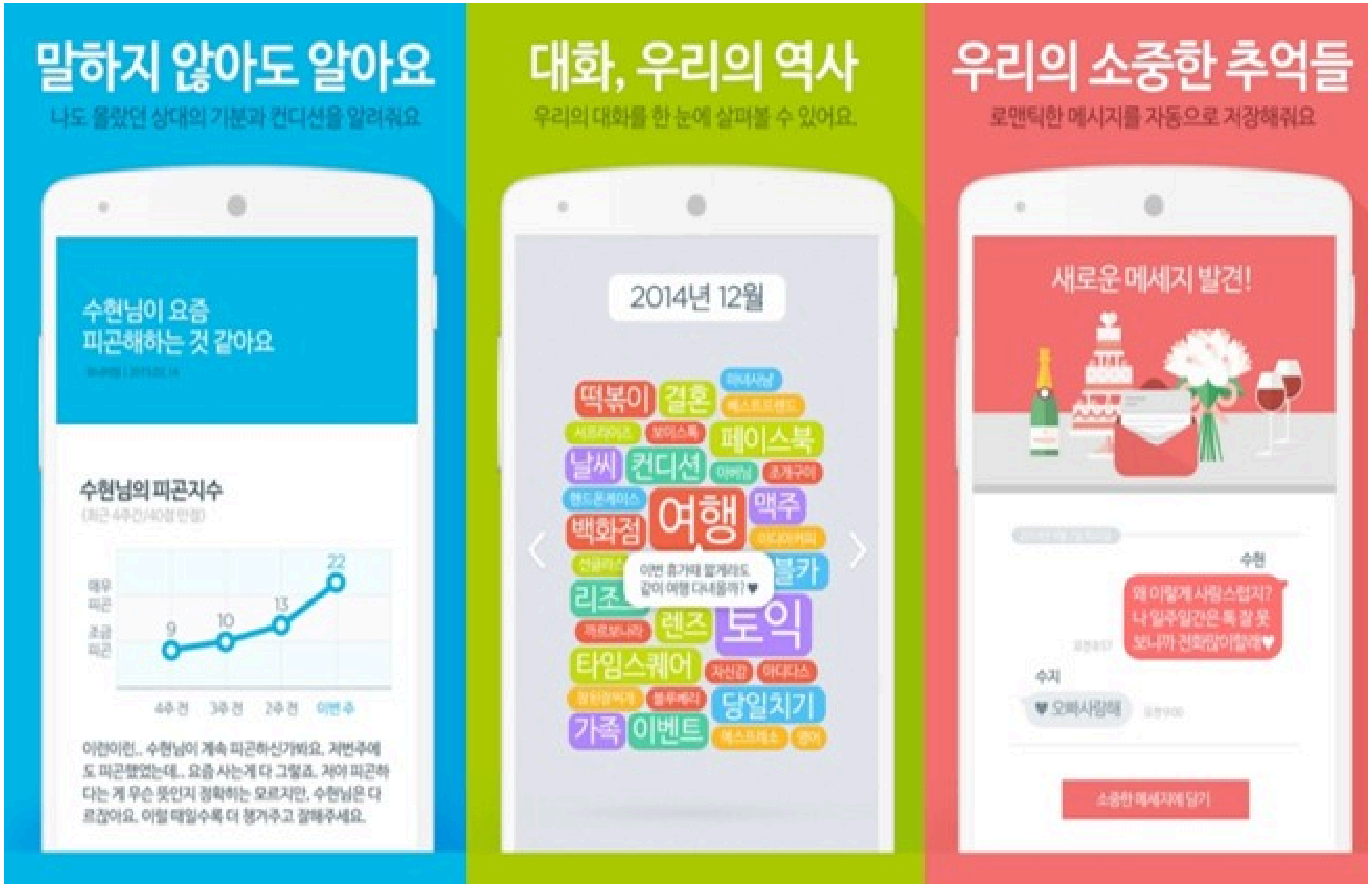
Text Mining Around Us

- Topic Detection/Trend Analysis
  - Analyze “buzz” words on SNS, web, news
  - E.g., “카카오 (kakao)”



Text Mining Around Us

- Relationship mining using messages between couples (e.g., Between app)
  - Trend analysis, topic discovery, and summarization



Text Mining Around Us

- Search Engine
- News Recommendation
- Question Answering
- Business Intelligence Tools
- ... and so on ...

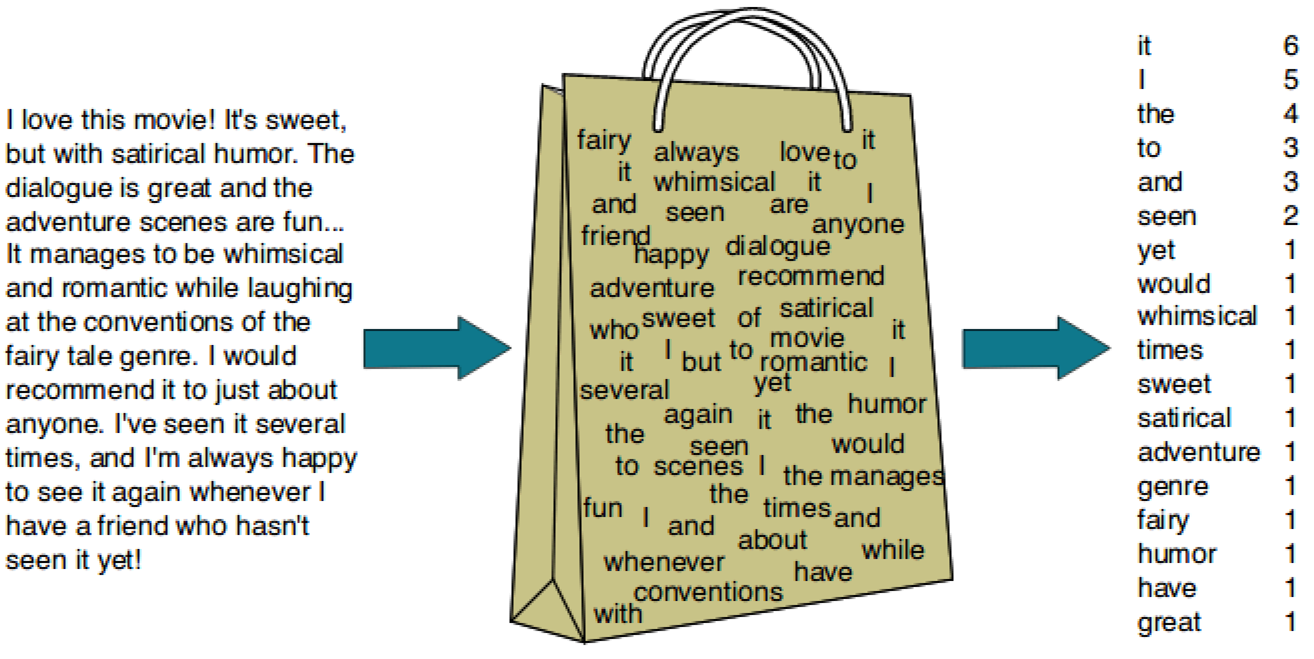
Corpus: Definition

- Simple, a collection of text data.
    - Usually but not necessarily *purposefully collected*
    - Usually but not necessarily *structured*
    - Usually but not necessarily *annotated*
- A.k.a. corpora

Bag-of-words (BOW) representation

How to represent a document or a corpus?

- Take all unique words from the corpus (a collection of documents)
- Count the frequency of occurrence and sort them in descending order



> 📝 BoW representation transforms unstructured text into a structured format.

Natural Language Processing (NLP)

How to extract "words" from a document?

We need **natural language processing (NLP)** techniques such as tokenization, Part-of-speech tagging, etc.

Text Processing Steps:

1. Parsing documents in sentences
2. Tokenization (상징화)
3. Part-of-speech (POS, 품사) tagging
4. Stopwords Removal (불용어 제거)
5. Stemming (대표형 변환)

1. Parsing documents in sentences

A fundamental step in many NLP tasks because syntactic analysis is typically performed at the sentence level.

Text

John likes to watch movies and Mary likes too.

John also likes to watch football game.

Sentences

S1: John likes to watch movies and Mary likes too.

S2: John also likes to watch football game.

- Use punctuation marks (`.` , `!` , `?`) at the end of sentences.
- Exceptions: `Ms. Kim` , `3.14` , `Y corp.` , ...

2. Tokenization (상징화)

Parsing text into the most basic unit of analysis (token)

S1: John likes to watch movies and Mary likes too.

S2: John also likes to watch football game.

S1: John / likes / to / watch / movies / and / Mary / likes / too

S2: John / also / likes / to / watch / football / game

- **English:** word-level tokenization is done using space between words English grammar requires spacing!

Korean tokenization is much more *difficult* than English tokenization!

- Spacing is not well observed, so spacing alone is incomplete.
  - E.g., 오늘기부니가조아, 정의상실, ...

- Complex morpheme (복합형태소) like `noun(명사)` + `postposition(조사)`
  - E.g., `커피` + `를`, `랑`, `말고`, `도`, `에서`, `와` ...

3. Part-of-speech (POS, 품사) tagging

Determining the appropriate POS for the identified tokens

S1: John likes to watch movies and Mary likes too .

S2: John also likes to watch football game .

S1: John/NNP likes/VBZ to/TO watch/VB movies/NNS and/CC Mary/NNP likes/VBZ too/RB ./.

S2: John/NNP also/RB likes/VBZ to/TO watch/VB football/NN game/NN ./.

- POS tagging is done based on the probability that a word exists in different locations depending on the POS
  - E.g., (English) ***this*** person vs. look at ***this***
  - E.g., (Korean) ***이*** 사람은 vs. 벼룩과 ***이***가

POS taggers for Korean

꼬꼬마 형태소 분석기

- 서울대학교 지능형 데이터 시스템 연구실 개발
- 기분석 사전 기반, 띄어쓰기 보정 기능 포함
- 구문 분석 기능 제공
- http://kkma.snu.ac.kr

한나눔 형태소 분석기

- KAIST Semantic Web Research Center (SWRC) 개발
- 용도별 사전, 맞춤법 교정 기능 포함
- 구문분석 기능 제공
- http://semanticweb.kaist.ac.kr/hannanum/

아리랑 한글 형태소 분석기

- Lucene 검색엔진의 한국어 색인어 추출을 위한 형태소 분석기
- http://cafe.naver.com/korlucene

KLT2000 형태소 분석기

- 국민대 한글공학-정보검색 연구실(강승식 교수) 개발
- http://nlp.kookmin.ac.kr/HAM/kor/index.html

Twitter 형태소 분석기

- 트위터에서 만든 한국어 처리기
- normalization, normalization, tokenization, phrase extraction 기능을 제공
- https://github.com/twitter/twitter-korean-text

mecab-ko

- 일본어 형태소 분석기 mecab 으로 세종 말뭉치를 학습하여 개발
- https://bitbucket.org/eunjeon/mecab-ko

KACTEIL-KMA

- 강원대학교 자연어처리 연구실 개발
- http://nlp.kangwon.ac.kr:8080/demo/K_ACTEIL-KMA/index.html

KOMORAN

- Java 로 개발된 경량화가 특징인 형태소 분석기
- https://github.com/shin285/KOMORAN

Comparison of POS taggers for Korean

Input: `아버지가방에들어가신다`

Output: `아버지가 방에 들어가신다` vs. `아버지 가방에 들어가신다`

Hannanum	Kkma	Komoran	Mecab	Twitter
아버지가방에 들어가 / N	아버지 / NNG	아버지가방에 들어가신다 / NNP	아버지 / NNG	아버지 / Noun
이 / J	가방 / NNG		가 / JKS	가방 / Noun
시ㄴ다 / E	에 / JKM		방 / NNG	에 / Josa
	들어가 / VV		에 / JKB	들어가신 / Verb
	시 / EPH		들어가 / VV	다 / Eomi
	ㄴ다 / EFN		신다 / EP+EC	

Input: 나는 밥을 먹는다

Hannanum	Kkma	Komoran	Mecab	Twitter
나 / N	나 / NP	나 / NP	나 / NP	나 / Noun
는 / J	는 / JX	는 / JX	는 / JX	는 / Josa
밥 / N	밥 / NNG	밥 / NNG	밥 / NNG	밥 / Noun
을 / J	을 / JKO	을 / JKO	을 / JKO	을 / Josa
먹 / P	먹 / VV	먹 / VV	먹 / VV	먹는 / Verb
는다 / E	는 / EPT	는다 / EC	는다 / EC	다 / Eomi
	다 / EFN			

Input: 하늘을 나는 자동차

Hannanum	Kkma	Komoran	Mecab	Twitter
하늘 / N	하늘 / NNG	하늘 / NNG	하늘 / NNG	하늘 / Noun
을 / J	을 / JKO	을 / JKO	을 / JKO	을 / Josa
나 / N	날 / VV	나 / NP	나 / NP	나 / Noun
는 / J	는 / ETD	는 / JX	는 / JX	는 / Josa
자동차 / N	자동차 / NNG	자동차 / NNG	자동차 / NNG	자동차 / Noun

Input: 아이폰 기다리다 지쳐 애플공홈에서 언락폰질러버렸다 6+128기가 실버

Hannanum	Kkma	Komoran	Mecab	Twitter
아이폰 / N	아이 / NNG	아이폰 / NNP	아이폰 / NNP	아이폰 / Noun
기다리 / P	폰 / NNG	기다리 / VV	기다리 / VV	기다리 / Verb
다 / E	기다리 / VV	다 / EC	다 / EC	다 / Eomi
지치 / P	다 / ECS	지치 / VV	지쳐 / VV+EC	지쳐 / Verb
어 / E	지치 / VV	어 / EC	애플 / NNP	애플 / Noun
애플공홈 / N	어 / ECS	애플 / NNP	공 / NNG	공홈 / Noun
에서 / J	애플 / NNP	공 / NNG	홈 / NNG	에서 / Josa
언락폰질러버렸다 / N	공 / NNG	홈 / NNG	에서 / JKB	언락폰 / Noun
6+ / N	홈 / NNG	에서 / JKB	언락 / NNG	질 / Verb
128기가실벅 / N	에서 / JKM	언 / NNG	폰 / NNG	러 / Eomi
	언락 / NNG	락 / NNG	질러버렸 / VV+EC+VX+EP	버렸 / Verb
	폰 / NNG	폰 / NNG	다 / EC	다 / Eomi
	질르 / VV	지르 / VV	6 / SN	6 / Number
	어 / ECS	어 / EC	+ / SY	+ / Punctuation
	버리 / VXV	버리 / VX	128 / SN	128 / Number

Kkma	Komoran	Mecab	Twitter
버리 / VXV	버리 / VX	128 / SN	128 / Number
었 / EPT	었 / EP	기 / NNG	기 / Noun
다 / ECS	다 / EC	가 / JKS	가 / Josa
6 / NR	6 / SN	실버 / NNP	실버 / Noun
+ / SW	+ / SW	ㅋ / UNKNOWN	ㅋ / KoreanParticle
128 / NR	128기가실벅 / NA		
기가 / NNG			
실버 / NNG			
ㅋ / UN			

4. Stopwords Removal (불용어 제거)

- Stopwords: words that do not carry any information
  - Words that perform most grammatical functions (article, conjunctions, ...)
  - Removing stopwords improves text mining performance

S1: John / likes / to / watch / movies / and / Mary / likes / too

S2: John / also / likes / to / watch / football / game

S1: John / likes / watch / movies / Mary / likes

S2: John / likes / watch / football / game

Stop-words list
- To
- And
- Too
- Also
- ...

- Language dependent
  - English: `a`, `about`, `above`, `after`, `also`, ...
  - Korean: `-습니다`, `-로서(써)`, `-를`, `그리고`, ...
- Corpus dependent
  - E.g., `movie` in a movie review corpus

5. Stemming (대표형 변환)

Words having the same meaning but existing in multiple forms for grammatical reasons.

Stemming is to transform a word into its minimal primitive form that has meaning.

S1: John / likes / watch / movies / Mary / likes

S2: John / likes / watch / football / game

S1: John / like / watch / movie / Mary / like

S2: John / like / watch / football / game

- **English**: `learn`, `learns`, `learning`, `learned` --> ***learn***
- **Korean**: `먹다`, `먹어서`, `먹는다`, `먹는`, `먹기`, `먹을` --> ***먹다***

Install packages for NLP

Required software and packages

- Java Development Kit (JDK)
- NLTK package for English
- KoNLPy package for Korean

Install Java (for MacOS users)

https://www.oracle.com/java/technologies/downloads/

Download Java installer (red box below).

Java 19 **Java 17**

Java SE Development Kit 19.0.1 downloads

Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications and components using the Java programming language.

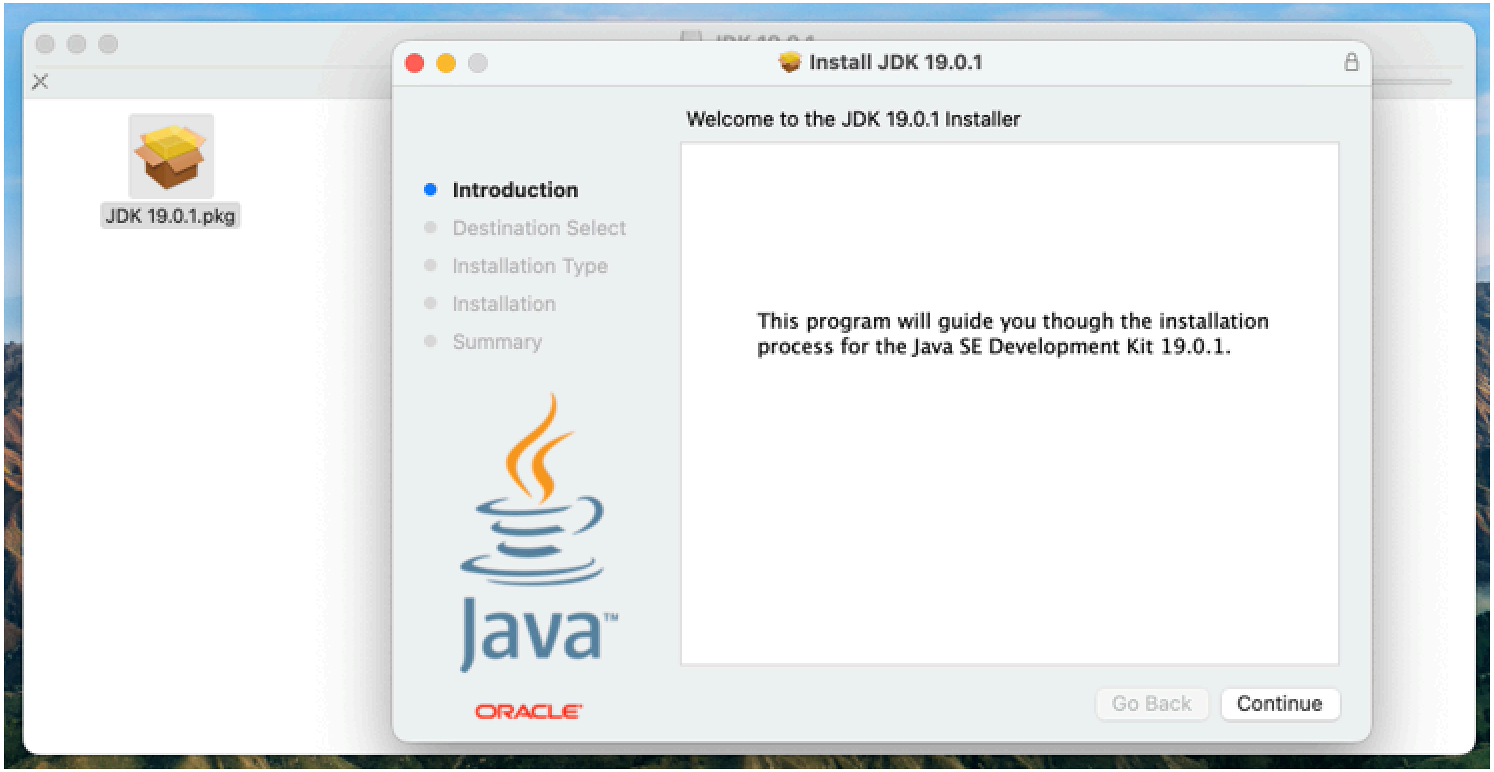
The JDK includes tools for developing and testing programs written in the Java programming language and running on the Java platform.

Linux **macOS** **Windows**

Choose one of the two according to your Mac Spec.

Product/file description		File size	Download
Arm 64 Compressed Archive		175.67 MB	https://download.oracle.com/java/19/latest/jdk-19_macos-aarch64_bin.tar.gz (sha256)
Arm 64 DMG Installer	Mac M1~	175.07 MB	https://download.oracle.com/java/19/latest/jdk-19_macos-aarch64_bin.dmg (sha256)
x64 Compressed Archive		177.54 MB	https://download.oracle.com/java/19/latest/jdk-19_macos-x64_bin.tar.gz (sha256)
x64 DMG Installer	Before M1	176.92 MB	https://download.oracle.com/java/19/latest/jdk-19_macos-x64_bin.dmg (sha256)

Run the installer to install Java (JDK)



Install Java (for Windows users)

https://www.oracle.com/java/technologies/downloads/

Download Java installer (red box below).

Java 19 **Java 17**

Java SE Development Kit 19.0.1 downloads

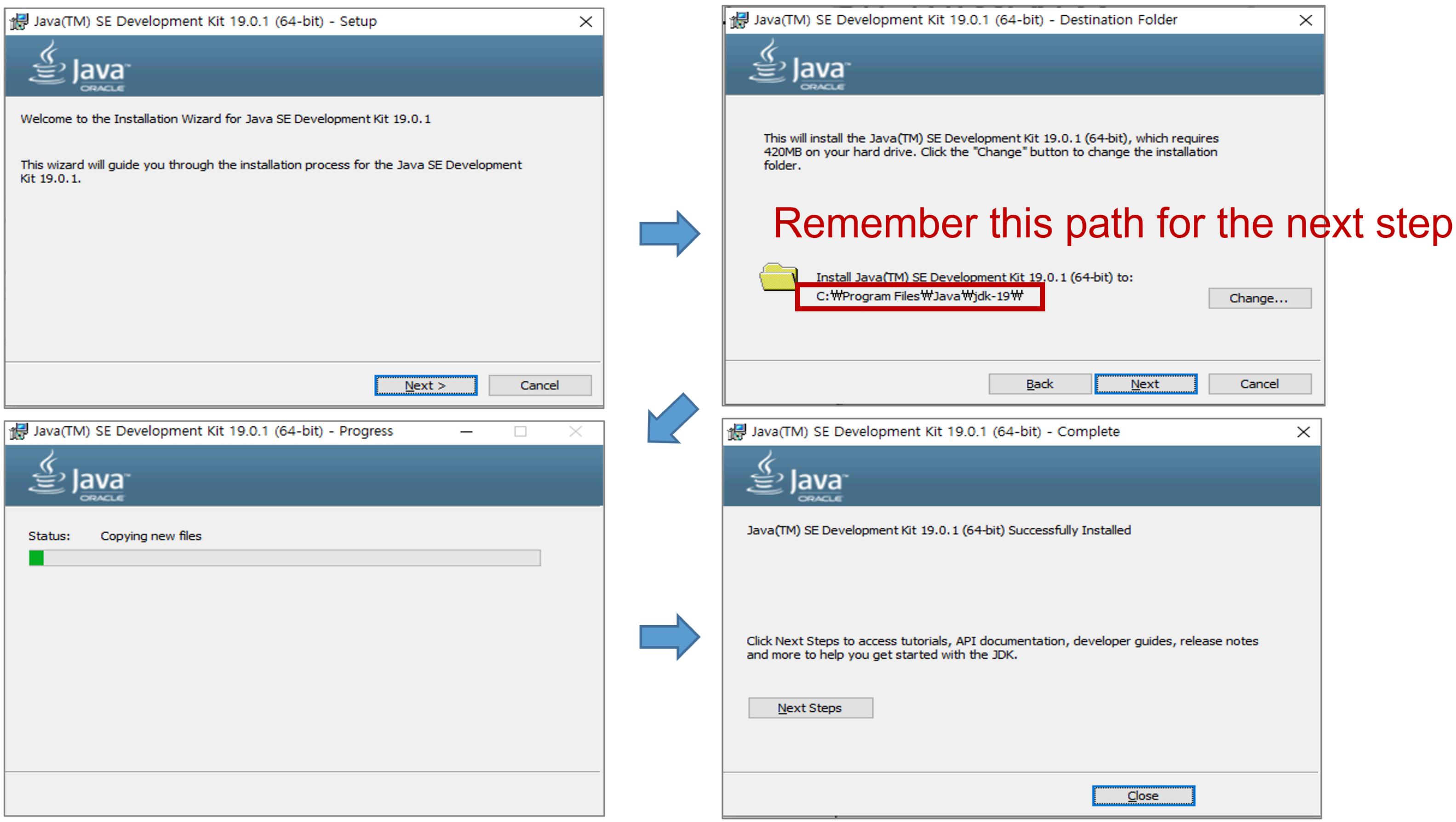
Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications and components using the Java programming language.

The JDK includes tools for developing and testing programs written in the Java programming language and running on the Java platform.

Linux **macOS** **Windows**

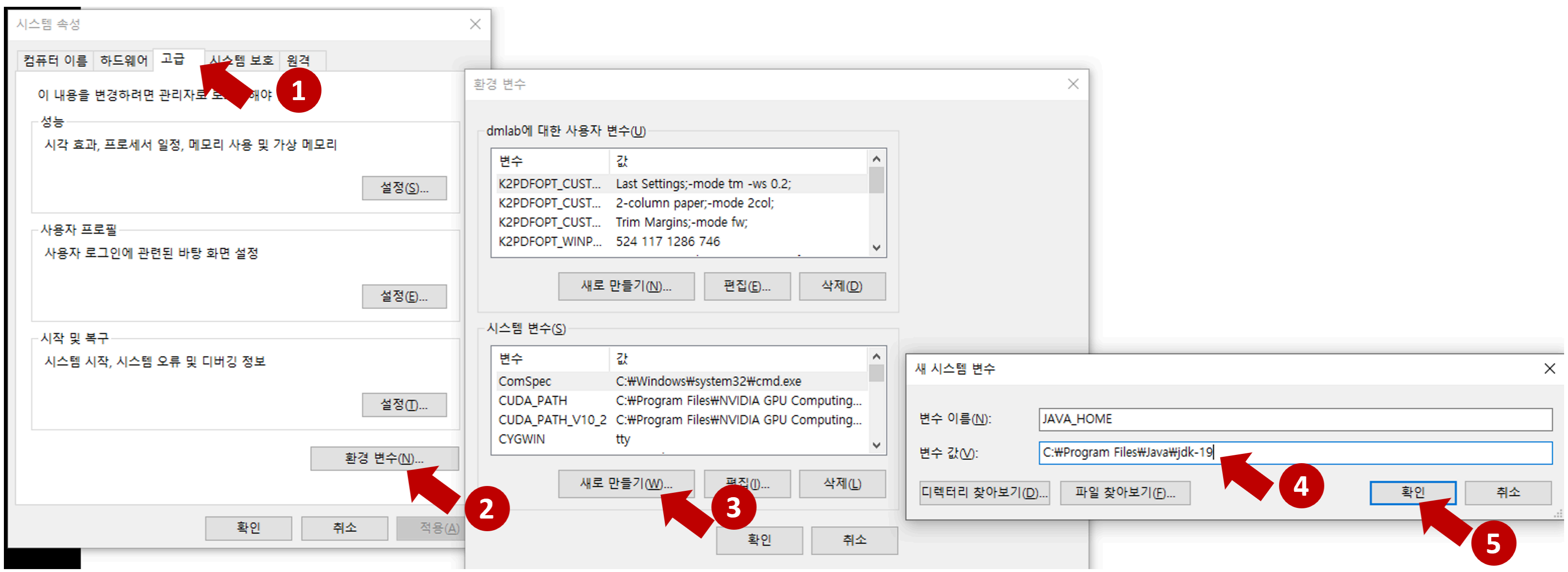
Product/file description	File size	Download
x64 Compressed Archive	179.13 MB	https://download.oracle.com/java/19/latest/jdk-19_windows-x64_bin.zip (sha256)
x64 Installer	158.91 MB	https://download.oracle.com/java/19/latest/jdk-19_windows-x64_bin.exe (sha256)
x64 MSI Installer	157.76 MB	https://download.oracle.com/java/19/latest/jdk-19_windows-x64_bin.msi (sha256)

Run the installer to install Java (JDK)



After installation, set `JAVA_HOME` environmental variable.

1. Right click My Computer and select Properties.
2. On the Advanced tab, select Environment Variables, and then edit `JAVA_HOME` to point to the installation path, for example, `C:\Program Files\Java\jdk-19` (in red box in the previous slide)



Install NLTK

0. Ensure you already installed Java on your computer.
1. Open anaconda prompt
2. Enter the following:

```
pip install nltk
```

3. Done!

Install KoNLPy

1. Open anaconda prompt
2. Enter the following:

```
pip install jpype1
pip install konlpy
```

3. Done!

Install wordcloud

1. Open anaconda prompt
2. Enter the following:

```
pip install wordcloud
```

3. Done!

FAQ: [WinError 5] Access denied (액세스가 거부되었습니다)

After running the `pip` command, an error appears:

```
ERROR: Could not install packages due to an OSError: [WinError 5] Access denied (액세스가 거부되었습니다)
```

```
Anaconda Prompt
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\[redacted]\.conda\envs\study\lib\site-packages (from requ
ests<3,>=2.21.0->tensorboard<2.13,>=2.12->tensorflow-intel==2.12.0->tensorflow) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in c:\users\[redacted]\.conda\envs\study\lib\site-packages (from requests<3,>=2.2
1.0->tensorboard<2.13,>=2.12->tensorflow-intel==2.12.0->tensorflow) (3.4)
Collecting MarkupSafe>=2.1.1
  Downloading MarkupSafe-2.1.2-cp310-cp310-win_amd64.whl (16 kB)
Requirement already satisfied: pyasn1<0.6.0,>=0.4.6 in c:\users\[redacted]\.conda\envs\study\lib\site-packages (from pyasn1-m
odules>=0.2.1->google-auth<3,>=1.6.3->tensorboard<2.13,>=2.12->tensorflow-intel==2.12.0->tensorflow) (0.5.0)
Requirement already satisfied: oauthlib>=3.0.0 in c:\users\[redacted]\.conda\envs\study\lib\site-packages (from requests-oaut
hlib>=0.7.0->google-auth-oauthlib<1.1,>=0.5->tensorboard<2.13,>=2.12->tensorflow-intel==2.12.0->tensorflow) (3.2.2)
Building wheels for collected packages: jax
  Building wheel for jax (pyproject.toml) ... done
  Created wheel for jax: filename=jax-0.4.10-py3-none-any.whl size=1480617 sha256=e700db02f67463b727dd9397e5eb8a2e6fb32a
5c549c91fb14924e3e3c9325b9
  Stored in directory: c:\users\[redacted]\appdata\local\pip\cache\wheels\2f\04\51\ebc9c5225f0a0df1e56c231c1f4c9b7afd3e024ebb
492eed99
Successfully built jax
Installing collected packages: libclang, flatbuffers, wrapt, termcolor, tensorflow-io-gcs-filesystem, tensorflow-estimat
or, tensorboard-data-server, protobuf, numpy, MarkupSafe, markdown, keras, grpcio, google-pasta, gast, astunparse, absl-
py, werkzeug, opt-einsum, ml-dtypes, jax, tensorboard, tensorflow-intel, tensorflow
  Attempting uninstall: numpy
    Found existing installation: numpy 1.24.3
    Uninstalling numpy-1.24.3:
      Successfully uninstalled numpy-1.24.3
ERROR: Could not install packages due to an OSError: [WinError 5] 액세스가 거부되었습니다: 'C:\\Users\\[redacted]\\.conda\\en
vs\\study\\Lib\\site-packages\\~umpy\\.libs\\libopenblas64__v0.3.21-gcc_10_3_0.dll'
Consider using the `--user` option or check the permissions.
```

Answer: Run the **Anaconda Prompt** as **Administrator** privilege.